# NeuralDocs ? AI & RAG Knowledge Base Guide

## 1. Overview of Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) empowers Large Language Models by grounding them with private document stores. Instead of relying solely on parametric training data, RAG retrieves precise contextual chunks matching user inquiries.

## 2. Architecture Breakdown

- Document Parsing: Ingestion pipeline parses incoming PDFs and separates content by page boundaries.
- Chunking & Overlap: Text is divided into semantic paragraphs with an overlap window to preserve continuity.
- Vector Embeddings: Numerical representations capturing semantic meaning are produced and stored inside ChromaDB.
- Semantic Querying: User questions are vectorized, and nearest cosine neighbors are returned as context.
- LLM Synthesis: Google Gemini processes the prompt along with the source chunks to formulate factual, cited answers.

## 3. Testing Guidelines for NeuralDocs Chatbot

- Upload this PDF using the 'Upload PDF' button in the sidebar.
- Ask questions like:
  * 'What is RAG according to this document?'
  * 'What database is used for storing vector embeddings?'
  * 'Explain the chunking and overlap mechanism.'